

Integrating societal preferences: Decentralization of the European energy network

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Abstract

Filler

Keywords:

European energy system, De-centralization, De-carbonization,
PyPSA-EUR, Energy systems modeling, Fairness in energy transitions

1. Introduction

In a world where the necessity of reducing the carbon emissions of our energy networks is no longer a question but a requirement careful and deliberate planning of the energy system of our future is needed. When the Paris Climate agreement was signed into action in 2015 it signaled the commitment of the United Nations to reducing their overall carbon footprint [1]. This common goal and understanding of the necessity of decarbonization was a crucial step towards a more sustainable world, and providing a common base understanding of what needs to be done, but also provides a massively complicated technological challenge that must be solved in order for the agreement to be fulfilled.

Somewhat unique to the issue of decarbonization in the European Union, is the fact that, although the Paris Climate Agreement signifies the common goal of decarbonization, The Union is still comprised of many different countries with many different political ideologies, laws, societal preferences and goals of their own. This means that although it has long been known that a shared and interconnected energy grid is the most efficient and economically preferable solution, as seen for example in [2], reality becomes a lot more complicated as the member nations also must have their own best interest in mind.

Historically the impact of the union being made up of self-centered actors has often been discarded when trying to construct the electricity grid of the future, if for no other reason than the fact that implementing social preferences and societal paradigms in strict mathematical models of the European energy system is a challenging problem [3]. These models that typically aim to reduce the overall cost of the system as the primary objective are often meant to be used as the basis for developing the future energy network and legislation. However using this least-cost optimization approach without considering social paradigms in the European Union tends to produce system models that, although being economically efficient at achieving decarbonization targets, that are intrinsically tied to the assumption that the EU has complete control over the member nations individual energy grids.

This approach of reducing the optimization problems to purely techno-economic problems that prioritize cost reduction over feasibility has helped achieve significant insights into decarbonization strategies for the European Union [4], but tends to lead to the energy system becoming heavily centralized [5]. Cen-

tralization in the context of energy systems refer to the majority of resources being pooled into a few areas that are favorable to the system in some way, be that installing a lot of wind turbines in a region with a lot of wind, maintaining energy production in regions close to where the energy is needed or any number of other mechanisms that result in a proportionally large investment into infrastructure in a single region. While centralization as a strategy is often the most cost efficient strategy it leads to a suite of other challenges.

Two such challenges are those of physical security and and political feasibility. With the recent surge in so-called hybrid attacks on European energy infrastructure[6], the focus on physical security has increased. If a system is overly centralized it means that a large amount of critical infrastructure is installed on a relatively small area. This makes the infrastructure an obvious target for such hybrid attacks as a single hostile action can impact more infrastructure and more people. As for the political feasibility, nations who end up as net energy importers become less self sufficient and increasingly reliant on the net exporting regions. This shift in power-balance heavily favors the net exporters and as such may face resistance from net importing nations that wish to remain at a high degree of self sufficiency.

The objective of this masters thesis is to incorporate societal paradigms and priorities in the form of decentralization into a macro-scale energy network model of the EU, and to analyze and expose the mechanisms behind decarbonization and its interactions with decentralization. The main hypothesis is that, by incorporating decentralization, the resulting networks will illuminate alternative transition pathways. This in tun may then help develop a workflow that by incorporating societal paradigms and reviewing their system impacts, will help close the gap between the models used to inform energy policy makers and the results of the proposed policies. Although this workflow and methodology may in time prove relevant for policy makers to understand, the target audience of this master’s thesis remains peers in the field of energy system modeling who may draw inspiration from the intersection between societal paradigms and strict mathematic modeling that this work represents. It is also for this reason that all the data and code used to produce the results which can be seen in this paper is openly available in a public repository under an open science license[7].

2. Nomenclature

Symbols

Symbol	Description
n	Node index (spatial unit)
t	Time snapshot index
g	Generator technology index
C	Decarbonization level (fraction of baseline emissions)
K	Decentralization parameter (capacity distribution factor)
$\lambda_{n,t}$	Marginal electricity price at node n and time t (EUR/MWh)
μ_C	Shadow price of CO ₂ emissions at decarbonization level C (EUR/tCO ₂)
p	Power output or capacity (MW)
e_g	Emission factor for generator technology g (tCO ₂ /MWh)
w_t	Temporal weighting factor for snapshot t
$L_{n,t}$	Electrical load (demand) at node n and time t (MWh)
ω_n	Normalized load-share weight for node n
R	Renewable generation capacity (MW)
r	Discount rate (per annum)
G	Gini coefficient of inequality

Abbreviations

Abbreviation	Definition
AC	Alternating Current
CAPEX	Capital Expenditure
CCGT	Combined Cycle Gas Turbine
EU	European Union
FOM	Fixed Operation and Maintenance
HPC	High Performance Computing
HVDC	High Voltage Direct Current
KKT	Karush-Kuhn-Tucker
LCOE	Levelized Cost of Electricity
OCGT	Open Cycle Gas Turbine
PHS	Pumped Hydro Storage
PyPSA	Python for Power System Analysis
SDRCL	Standard Deviation of Nodal Renewable Capacity per Load

3. Methodology

3.1. Modeling framework

For the purpose of structuring the modeling a Snakemake workflow was constructed. Snakemake is a workflow management system that allows for modular and reproducible data analysis [8]. The workflow is divided into multiple steps or "rules" that when chained together ensures that the data is processed in the correct order. The complete workflow used to produce the results in this master's thesis is openly available in a public repository [9] under a GNU General Public License [10].

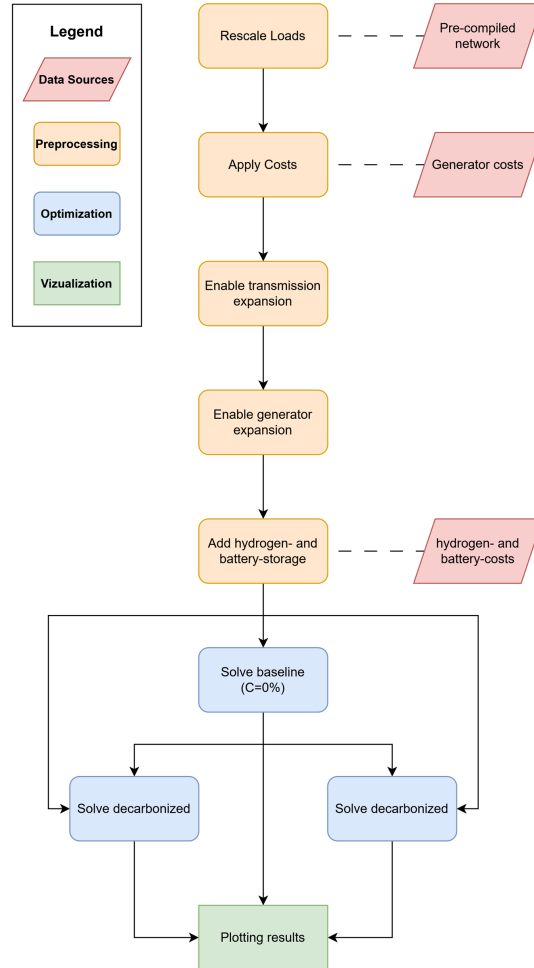


Figure 1: Map of Snakemake Workflow

Seen above in Figure 1 is a map of the rules and data-input used in the workflow divided into three main sections: Preprocessing, optimization and vizualization/postprocessing. The following sections will go further into detail on the diferent sections of the workflow, excluding the vizualizations and postprocessing which are primarily used for producing figures and tables for this master’s thesis.

3.2. Precompiled network model

Many energy modeling frameworks and pieces of commercially available software for optimizing nodal-networks exist. The work represented by this paper utilizes the open-source framework PyPSA (Python for power system analysis [11]). The reasoning for this is twofold: Firstly PyPSA is completely free and open source to use. This encourages open and collaborative science as well as transparency in the philosophy behind the energy system modeling. Secondly PyPSA consists of two separate parts. A modeling workflow based in the Python programming language, and a large database containing relevant open source data for constructing models, particularly of the European Union. This dataset is referred to as PyPSA-Eur[12] and is invaluable in its quality of data for building macro energy system models in the PyPSA workflow. This masters thesis utilizes both parts of the PyPSA environment. Acting as a baseline for all of the network topology for the European Union is a pre-compiled network developed with PyPSA-EUR using the default settings as published in the Zenodo repository [13].

The simple philosophy behind the PyPSA models is that a network is perceived as multiple interconnected nodes. These nodes and connections are then given a set of constraints that define the feasible solution space. Examples of constraints could be that each node is assigned an electrical load that must be serviced based on historical consumption, capacity factor time-series based on solar- and wind-patterns, or maximum or minimum capacity constraints like a maximum allowable power flow from one node to another representing the transmission infrastructure in the network. The network used in this master’s thesis is a 37 node network representing the 25 (excluding the island states of Malta and Cyprus) member nations of the EU. Additionally Switzerland, Norway, Bosnia-Herzegovina, Albania, Montenegro, North Macedonia, The United Kingdom and Serbia are included as their electrical infrastructure is connected to that of the EU. The following nations are also

represented by more than one node as they have large populations on islands: Denmark, Spain, Great Britain and Italy.

These nodes are then interconnected with HVDC and AC links and lines based on real-world connections. A graphical overview of the network can be seen below in Figure 3.2:

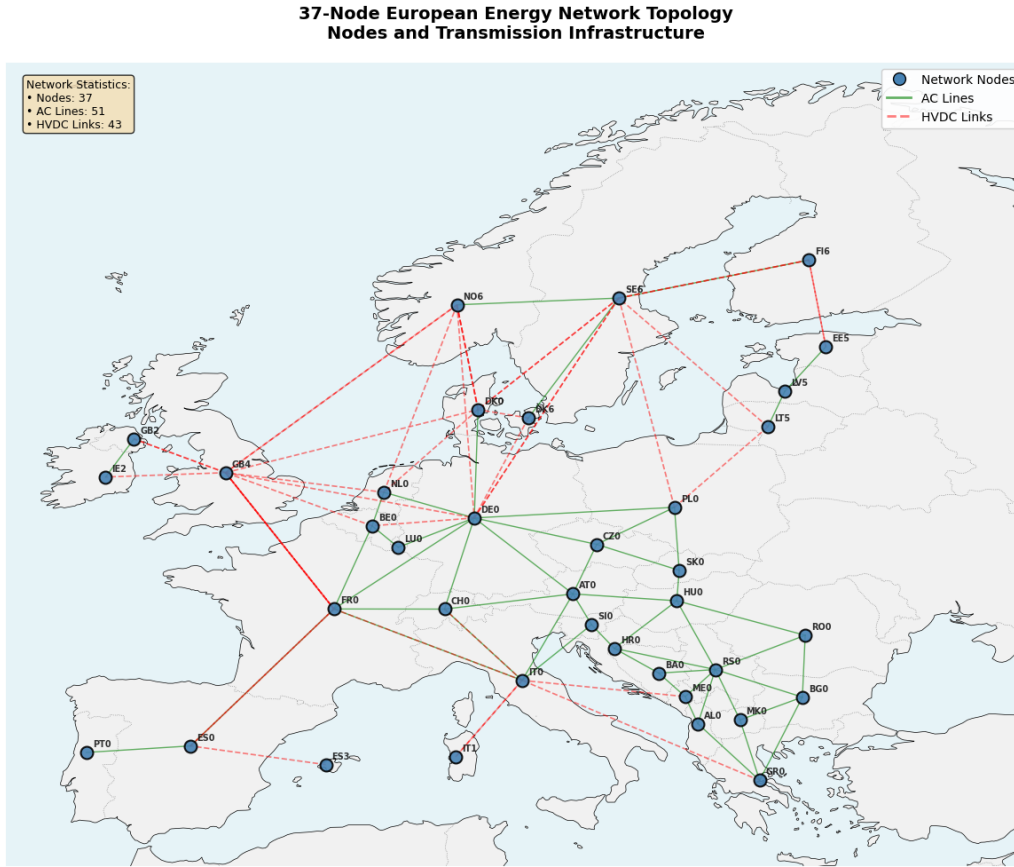


Figure 2: Overview of the 37-node European energy network.

Should i add something about spatial and temporal resolution and clustering?

3.3. Load rescaling

The precompiled network models comes preloaded with electrical demand time series for each of the nodes from the ENTSOE-E transparency platform

[14]. These hourly demand time series are, however, in the case of the pre-compiled model from the year 2013. In order to ensure a more representative demand profile for the nodes the loads have been rescaled to projected values for 2025. Each node is assigned a target electricity consumption value based on figures from Carbon-Free Europe [15]. These are then used together with the load values from 2013 to calculate a nodal scaling factor for each node as seen in equation (1) below:

$$\text{scaling factor}_n = \frac{\text{target Load}_n}{2013 \text{ Load}_n} \quad \forall n \quad (1)$$

Here n denotes the set of all nodes. These nodal scaling factors are then used as weights on the hourly load time series for each node to produce a rescaled load time series for each node as seen in equation (2) below:

$$\text{rescaled Load}_{n,t} = \text{scaling factor}_n \cdot 2013 \text{ Load}_{n,t} \quad \forall n, t \quad (2)$$

Where t denotes the set of all time snapshots. This way the total load is scaled to the projected value but the general shape of the load time series is preserved. An example of a nodal load time series before and after rescaling can be seen below in Figure 3:

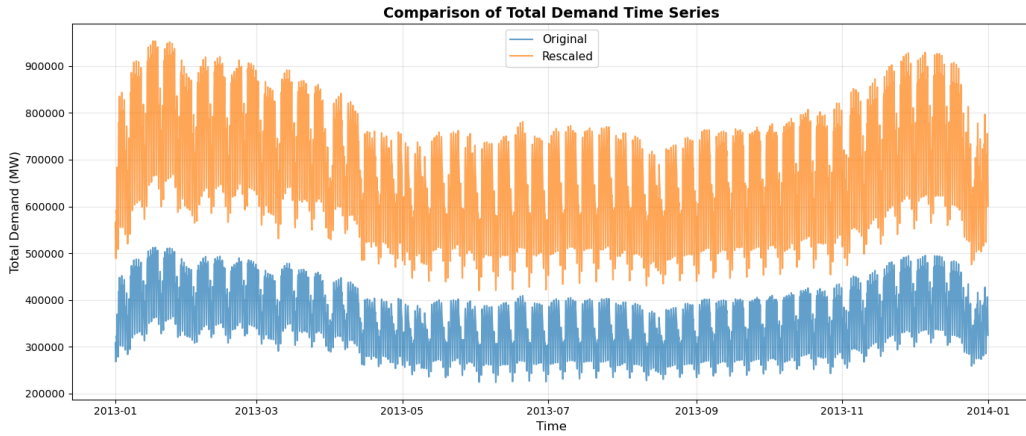


Figure 3: Example of load rescaling for node representing Denmark.

Here it can be seen that the shape of the load time series has been preserved while the total load has been rescaled to match the projected values.

3.4. Generator Cost Modeling

Although the precompiled network comes with some cost data already defined, there are some technologies that do not have cost data defined. In order to be able to optimize the network fully this must be rectified. Table 1 below shows the cost data used for the different generation and storage technologies in the network.

Table 1: Cost assumptions for generation technologies [16] [17] [18]

Carrier	CAPEX (EUR/MW)	Lifetime (years)	Discount rate	FOM (abs) (EUR/MW/a)	FOM (rate) (per a)	Notes
CCGT	830,000	25	0.07	27,800	–	DEA 2030
OCGT	435,000	25	0.07	7,745	–	DEA 2030
Coal	1,860,000	30	0.07	30,355	–	DEA 2030
Lignite	1,860,000	30	0.07	30,355	–	Proxied as coal CAPEX
Oil	343,000	25	0.07	8,448	–	DEA 2030
Biomass	3,300,000	25	0.07	96,000	–	DEA 2030
Geothermal	4,290,000	20	0.07	–	0.02	EGS ~2% FOM
Nuclear	5,870,000	60	0.07	–	0.02	NEA 2020 ~2% FOM

PyPSA, stores generator cost parameters as annualized total ownership costs. Due to differences in the way different sources report costs, some conversion is necessary in order to convert the reported costs into the format used by PyPSA. Firstly, for sources that report the FOM (Fixed Operation and Maintenance) costs as an absolute value the total annualized cost is calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} + \text{FOM (abs)} \quad (3)$$

Where the Annuity Factor is calculated as:

$$\text{Annuity Factor} = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (4)$$

With r being the discount rate and n being the lifetime of the technology in years. For sources that report FOM as an annual rate the annualized cost is calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} + \text{CAPEX} \cdot \text{FOM (rate)} \quad (5)$$

And finally for sources that report neither FOM (abs) nor FOM (rate) the annualized cost is simply calculated as calculated as:

$$\text{Annualized Cost} = \text{CAPEX} \cdot \text{Annuity Factor} \quad (6)$$

Performing these operations ensures that all the cost data is formatted the same when interfacing with the PyPSA model. The full cost assumptions for the different types of generators can be found in Appendix A: Generator cost assumptions.

3.5. Transmission- and generator-expansion

In order to find the optimal network the transmission infrastructure as well as select generator types must be allowed to vary their capacity. For transmission infrastructure the line capacity was constrained as seen in equation (7) below:

$$p_{\text{existing}} \leq p_{\text{line}} \leq p_{\text{line, max}} \quad (7)$$

Where p_{existing} is the existing capacity of the line, p_{line} is the optimized capacity of the line and $p_{\text{line, max}}$ is the maximum allowable capacity of the line. The minimum capacity is set to the pre-existing capacity as it is assumed that the existing transmission infrastructure will not be decommissioned due to the trouble of digging up existing lines. In order to avoid transmission bottlenecks, and find the optimal solution of installed generation capacity in the nodes, the maximum capacity was set to an arbitrarily high value.

A similar approach was used for the *fossil-based* generators in the system. These generators are allowed to decommission in order to meet the decarbonization targets, but are not allowed to expand their capacity beyond the existing capacity as can be seen in equation (8) below:

$$p_{\text{nom}} \leq p_{\text{existing}} \quad (8)$$

Most of the renewable generators are allowed to expand their capacity between 0 and an arbitrarily high value as these are the capacities that will later

be constrained by the decentralization measures. However some renewable generators like nuclear, geothermal and biomass are, constrained to "only" expand a factor 100 beyond their existing capacity into the nodes where they are already present. This is primarily done as calculating the maximum installable capacity for these technologies is a large task in itself and is outside the scope of this master's thesis. The freely expandable renewable technologies in the system are therefore solar PV, onshore wind and offshore wind which are constrained as seen in equation (9) below:

$$0 \leq p_{\text{nom}} \quad (9)$$

This does not however mean that there are no limits to the nodal capacities of these renewables, but rather that these are "soft" limits constrained by the capacity factor time series for solar and wind that determine the amount of wind and sun available at each node at each time snapshot.

3.6. Energy storage infrastructure

Finally, the last step before the system is ready to be optimized, is to implement the energy storage infrastructure. The pre-compiled network comes with a few different types of storage already implemented: Pumped-Hydro Storage (PHS), and Regular (reservoir) hydro storage. Historically hydro based storage in the European Union has been the primary method of energy storage and as a result most areas capable having hydro based storage installed have already been utilized. As a result of this the hydro based storage technologies are not allowed to change their capacity in the optimization and are constrained as seen in equation (10) below:

$$p_{\text{nom}} = p_{\text{existing}} \quad (10)$$

However as the system decarbonizes and the amount of dispatchable fossil-based generation decreases it is increasingly necessary to have large amounts of energy storage capacity installed. In order to properly model this two additional storage technologies, namely Battery storage and hydrogen storage has been added to the network. Battery storage consists of batteries and inverters and hydrogen storage consists of a hydrogen tank, an electrolyzer as well as a fuel cell. When adding the additional storage technologies their costs have been calculated in a manner similar to that of the generators as

seen in section 3.4 above. The cost assumptions for battery and hydrogen storage can be seen below in Table 2:

Table 2: Storage technology cost assumptions used in the optimization model [19].

Component	Type	CAPEX (EUR/unit)	Lifetime (years)	Efficiency	Annualized Cost (EUR/unit/a)
Battery	Energy (Cells)	142 kWh	25	—	12.2k MWh/yr
	Power (Inverter)	160 kW	25	0.96	13.7k MW/yr
Hydrogen	Energy (Tank)	57 kWh	25	—	4.9k MWh/yr
	Power (Electrolyser)	600 kW	30	0.66	48.4k MW/yr
	Power (Fuel Cell)	1100 kW	10	0.50	156.6k MW/yr

Here it is important to note that the storage units include efficiencies for charging and discharging. This efficiency is incorporated in optimization model as it has a large impact on system cost, which was annualized as per equations in section 3.4 above. As for the complete cost assumptions for the different storage components these can be found in Appendix B: Storage cost assumptions.

3.7. Optimization strategy

Optimizing complex energy networks with many variables and constraints is computationally very intensive. For this reason the optimizations performed for this master’s thesis have exclusively been performed on a High Performance Computing (HPC) environment. More specifically the UCloud HPC operated by SDU [20]. The optimizations were typically run on a set of 4 machines, each with an Intel Xeon Gold 6130 CPU and took on average 12 core-hours apiece to achieve an optimal solution. The optimization algorithm was chosen as the Barrier method in the commercial optimizer by Gurobi, chosen primarily for its numerical stability [21].

3.8. Baseline system

As seen in Figure 1 above, in order to be able to analyze the impacts of varying decarbonization constraints and decentralization measures first a baseline system that is subjected to neither decarbonization nor decentralization constraints must be optimized. This system, which will be used for comparison

is referred to as the *baseline* system. The optimization of this system is performed based on the objective function and load-balance, and is a *least cost* system, meaning that the total system cost is minimized.

In order to optimize the networks first an objective function must be defined. An objective function is, as the name suggests, the function which when minimized or maximized leads to the objective of the optimization to be achieved. The objective function for finding the cheapest possible electrical system is defined as minimizing the sum of all the costs:

Equation:

$$\min \left(\sum_n \text{generation costs} + \text{storage costs} + \text{transmission costs} + \sum_{n,t} \text{variable costs} \right) \quad (11)$$

Here n denotes the set of all nodes, and t denotes the set of all snapshots during the timeframe. However in order to construct a feasible solution space the objective function must be defined by a set of bounding constraints. Initially an equality constraint is defined so that, for each snapshot, all nodes must have their electricity demand met by a combination of energy production and nodal balancing:

$$\text{generation}_{n,t} + \text{balance}_{n,t} = \text{demand}_{n,t} \quad \longleftrightarrow \quad \lambda_{n,t} \quad \forall n, t \quad (12)$$

Here λ is known as the dual-variable or the Karush-Kuhn-Tucker (KKT) multiplier associated with the constraint. This variable can be interpreted as the shadow price, or marginal price of electricity in node n at time t . This marginal price represents the cost of supplying one additional unit of electricity to a node and as such is set by the most expensive generator needed to meet demand in that node at that time[22].

Should i explain how the KKT multipliers work here?

Optimizing the network with, only in regards to the objective function (11) and the load balance constraint (12) is what produces the baseline system that will be used for comparrison in the following sections.

3.9. Decarbonized systems

Once the baseline system has been optimized the a set of new networks at different levels of decarbonization can be defined and optimzed. In addition

to the objective function (11) and the load balance (12) these systems are also constrained by a decarbonization constraint as seen in equation (13) below:

$$\text{total emissions}_C \leq (1 - C \cdot \text{baseline emissions}) \quad \longleftrightarrow \quad \mu_C \quad (13)$$

Where C is the level of decarbonization as a fraction of the baseline emissions (the total emissions from the baseline system) and μ is the KKT multiplier and can be interpreted as the price of emitting an additional unit of CO₂ in the system at decarbonization level C . The baseline emissions are calculated based on the emission factors of each generator technology and their respective power output as seen in equation (14) below:

$$\text{total emissions} = \sum_t \sum_n \left(\sum_g p_{g,n,t} \cdot e_g \right) \cdot w_t \quad (14)$$

Here p is the power output of generator g in node n at time t , e is the emission factor of generator g and w is the weighting factor for snapshot t .

Based on previous experience with optimising energy systems the following levels of C were chosen for the reference systems:

$$C \in \{0\%, 10\%, 20\%, 30\%, 40\%, 50\%, 60\%, 70\%, 80\%, 90\%, 93\%, 96\%, 99\%, 100\%\} \quad (15)$$

This was done in order to achieve a wide range of decarbonization, while increasing resolution at higher levels of decarbonisation where the system tends to change more rapidly. The outcome of the the decarbonization workflow is therefore a set of 14 different networks (excluding the baseline), each at a different level of decarbonization defined as a fraction of the baseline emissions.

3.10. Decentralized systems

The first step in creating a set of decentralized systems, at each level of decarbonization, is to define a measure of decentralization. Some papers like "Modeling all alternative solutions for highly renewable energy systems" by Pedersen et al. [23] use the *Gini Coefficient* of inequality of installed renewable capacity as a measure of decentralization:

$$G = \frac{n + 1 - 2 \cdot \frac{\sum_{i=1}^n C_i}{C_n}}{n} \quad (16)$$

Here n is the number of nodes, C_i is the cumulative installed capacity at node i when the nodes are sorted in ascending order of installed capacity and C_n is the total installed capacity in the system. This, however comes with a number of issues. primary among them is that fact that the Gini coefficient is non-linear and as such cannot directly be used as a constraint as this will lead to non-convex solution spaces and local minima. While it is possible to work around this by using a sampling based approach and then calculating the Gini coefficient post optimization, this approach is difficult to understand and implement. As a result the decentralization of the systems was constrained as seen in equation (17) below:

$$\frac{1}{K} \cdot \omega_n \cdot R_{\text{total}} \leq R_{\text{cap},n} \leq K \cdot \omega_n \cdot R_{\text{total}} \quad \forall n \quad (17)$$

Where $\omega_n = \frac{L_n}{\sum_{n'} L_{n'}}$ is the normalized load-share weight for node n , R_{total} is the total renewable capacity across all nodes, $R_{\text{cap},n}$ is the installed renewable capacity at node n , and K is the decentralization parameter. This way each node is given a minimum renewable capacity as a fraction of the total renewable capacity in the system based on their share of the total load. Simultaneously each node is also assigned a maximum renewable capacity as a factor of the total renewable capacity in the system based on their share of the total load. If, for examle $K = 3$ that means that each node is allowed to "outperform" their load-share based renewable capacity by a factor of 3, while also being required to install at least one third of their load-share based renewable capacity. By then gradually having K approach 1 the system is forced to decentralized until at $K = 1$ each node is required to install renewable capacity exactly equal to their load-share based renewable capacity, which leads to the system becoming increasingly decentralized.

Based on previous preliminary experiments following levels of K were chosen for the decentralized systems:

$$K \in \{8, 7, 6, 5, 4, 3, 2, 1\} \quad (18)$$

This approach has the benifit of being linear, but also avoids another pitfall when implementing decenralization in systems that also are subjected to decarbonization constraints. Both these constraints, in some way, interact with

the amount of renewable capacity installed in the nodes of the network. If one was to simply constrain the amount of installed renewable capacity in each node without taking into account the total amount of renewable capacity in the system, it could lead to the system installing more renewable capacity than needed to meet the decarbonization constraint. This interaction would then lead to the system becoming overly decarbonized and difficult to compare to other systems.

The final outcome of the decentralization workflow is therefore a set of 112 different networks, each at a different level of decarbonization and decentralization. Before proceeding to the results of the optimization workflow it is useful to introduce a more intuitive measure of the decentralization of a given system as the K parameter is somewhat abstract. A more intuitive measure of decentralization is the *Standard Deviation of Nodal Renewable Capacity per Load* defined as:

$$\text{SDRCL} = \sqrt{\frac{1}{n} \sum_n \left(\frac{R_{\text{cap},n}}{L_n} - \mu \right)^2} \quad \text{where} \quad \mu = \frac{1}{n} \sum_n \frac{R_{\text{cap},n}}{L_n} \quad (19)$$

Here $R_{\text{cap},n}$ is the installed renewable capacity at node n , L_n is the load at node n and n is the number of nodes. This measure represents the standard deviation of the ratio between installed renewable capacity and load at each node. A low value of SDRCL indicates that the renewable capacity is more evenly distributed across the nodes relative to their load, while a high value indicates a more uneven distribution. By then comparing the SDRCL values for each of the decentralized systems to the network at the same level of decarbonization but without decentralization constraints, it is possible to get a more intuitive measure of how decentralized a given system is and how successful the decentralization constraints were.

4. Results

Introduction to results

4.1. Least-cost systems

Maps and figures showing a few least-cost networks, their optimized topology and key figures like system cost and marginal prices.

4.2. Zero cost condition, brownfield systems and scarcity-rents

When reviewing the electricity prices associated with the optimized networks, it is of interest to compare two different cost measures: The mean marginal demand-weighted price and the levelized cost of electricity (LCOE). The mean marginal demand-weighted price is defined as:

$$\text{Mean Marginal Demand-Weighted Price} = \frac{\sum_{t \in T} \sum_n (L_{n,t} \times \lambda_t)}{\sum_{t \in T} \sum_n L_{n,t}} \quad (20)$$

For time t , Nodes n , Loads L and marginal price λ . This cost measure represents the average price that consumers pay for electricity over the entire time period, weighted by their demand. The other cost measure, the levelized cost of electricity (LCOE) is defined as:

$$\text{LCOE} = \frac{\sum_{t \in T} \sum_g (\text{Capital Costs}_{g,t} + \text{Marginal Costs}_{g,t})}{\sum_{t \in T} \sum_n L_{n,t}} \quad (21)$$

For generator technologies g . LCOE represents the average cost of producing electricity over the lifetime of the system.

Seen below in Figure 4 is a plot comparing the mean levelized cost of electricity for the different levels of decarbonization, compared to the mean marginal demand-weighted price:

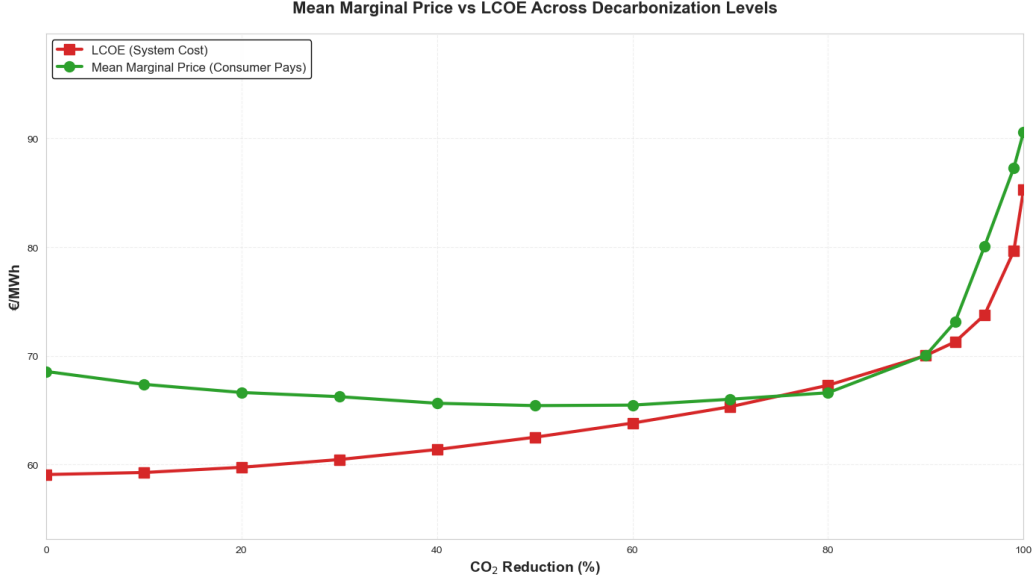


Figure 4: Comparison of mean LCOE and mean marginal price

This plot shows two different interesting results. Firstly it can be seen that as the system begins to decarbonize, initially the mean marginal demand weighted price *decreases* before increasing dramatically at high levels of decarbonization. Secondly it can be seen that for a majority of decarbonization levels a significant gap between the two cost measures is present.

Traditionally optimized energy system models such as the ones produced by PyPSA must adhere to the zero-profit condition [24]:

$$\sum_{t \in T} \sum_n (L_{n,t} \times \lambda_t) = \sum_{t \in T} \sum_g (\text{Capital Costs}_{g,t} + \text{Marginal Costs}_{g,t}) \quad (22)$$

This equation states that for all times the demand-weighted marginal price in all nodes must be equal to the sum of the capital and marginal costs across all technologies and times. Or, in other words: The revenue must be equal to the expenditure. However, critically, the zero-profit condition applies to long-run competitive equilibrium subject to the following assumptions:

- All capacity is free to expand and contract in order to achieve optimality.

- No artificially imposed constraints on the system.
- Steady state market conditions.

This clashes with the modelling methodology in several ways. The optimized model is a *Brownfield* system, meaning that even before optimization some fossil generation capacity exists in the system. As stated in the modelling section XXX, fossil generation capacity is allowed to contract but not to expand as constrained by:

$$p_{\text{nom}} \leq p_{\text{nom, max}} = p_{\text{nom, baseline}} \quad (23)$$

This means that the zero profit condition does not apply to optimization performed in this thesis, which leads to the difference between levelized cost and mean marginal price, also known as scarcity rent, as seen in Figure 4. The scarcity rents manifest when the optimizer ideally would increase capacity to meet demand, but is constrained by a binding capacity constraint.

Imagine for example that a fossil fuel-based generator is producing electricity at a lower marginal cost than the current market price. If the fossil based generator was allowed to expand, this would naturally attract additional fossil based generation to the market, until it becomes saturated and the mean marginal demand weighted price decreases to the point where the fossil generator is no longer profitable. However, as the fossil generator is not allowed to expand capacity, it can continue to produce electricity at a profit, leading to the scarcity rents seen in the system. This is the primary driving mechanism behind the gap between LCOE and mean marginal demand weighted prices seen in Figure 4.

This mechanism also applies to non-extendable renewable generation capacity, like nuclear, geothermal and biomass, as well as non-extendable storage like pumped hydro-storage. Pictured below, in Figure 5 is the evolution of the contribution to the total scarcity rent from the different technologies not constrained by the zero-profit condition:

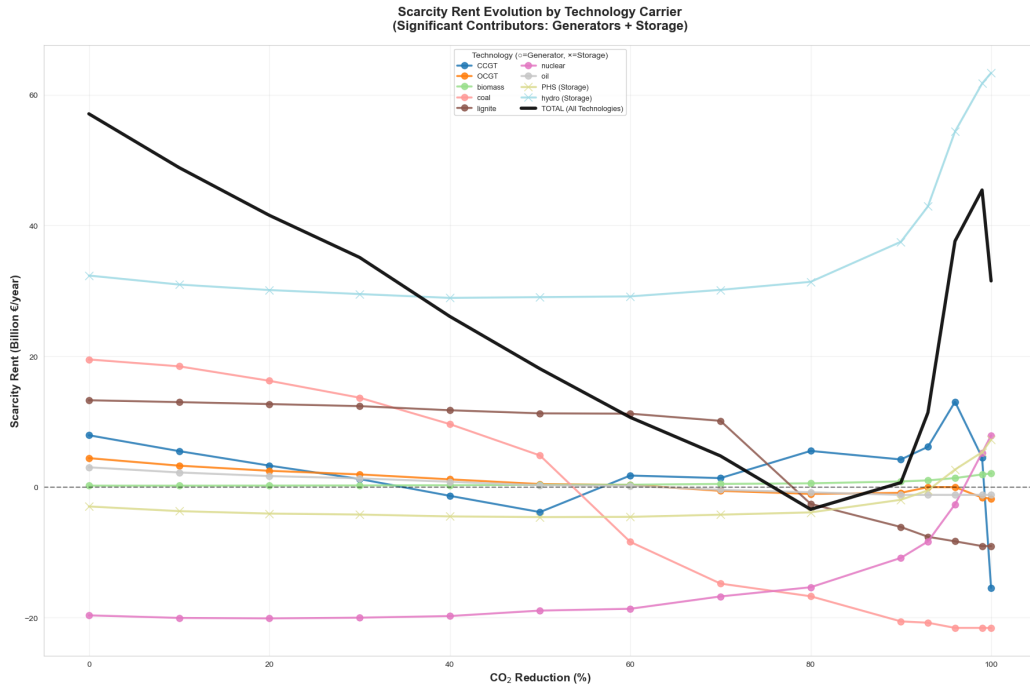


Figure 5: Scarcity rent contributions by technology

Here it can be seen that at low levels of decarbonization the majority of the contribution to the total scarcity rent comes from the fossil-fuel based generators. As the system decarbonizes the amount of fossil fuel generation is reduced in order to meet the decarbonization constraint and as such the contribution to the scarcity rent decreases. However at around 80% decarbonization where the vast majority of the fossil based generation has been phased out, the main contributors become the dispatchable renewable generators, such as nuclear, and storage units needed for balancing. This is illustrated in the modified version of Figure 4 below:

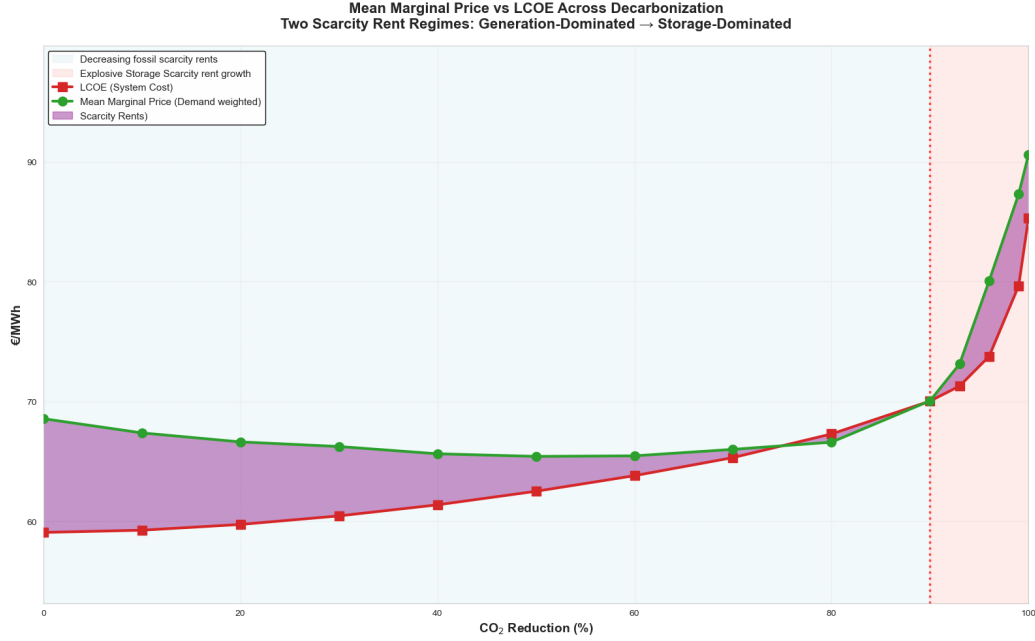


Figure 6: Transition between main scarcity rent contributors

As a side note, a negative scarcity rent indicates that the technology in question is running at a loss, meaning that the revenue from selling electricity is not sufficient to cover the capital and marginal costs of the technology. One might think that such a technology would not be included in the optimized system, however as the optimization is performed on a brownfield system with existing capacity, it may be optimal to keep such a technology running at a loss if the overall system cost is reduced by doing so.

As for the initial decrease in mean marginal demand weighted price. This can be attributed to something called the "merit-order effect": When the decarbonization constraint of fossil fuel energy generation tightens, fossil-fuel based energy production is replaced by renewable generation. As some forms of renewable generation have very low marginal costs the mean marginal demand weighted price decreases as these low marginal cost generators set the market price in more snapshots. However the total system cost increases as the newly installed generation- and storage-capacity has capital costs that contribute to the overall increase in system cost as seen in Figure 7 below:

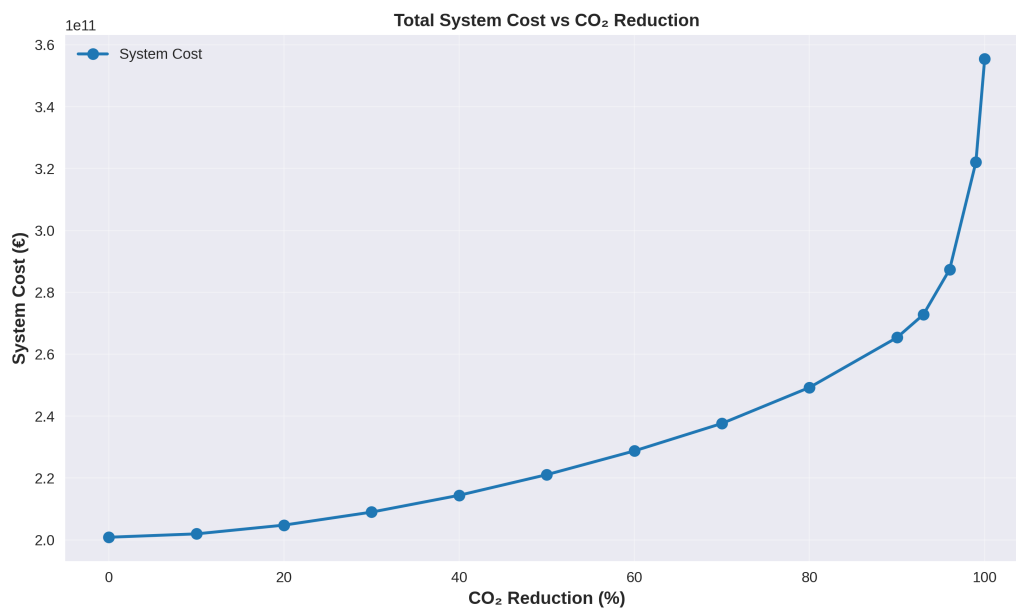


Figure 7: Total system cost in least-cost systems

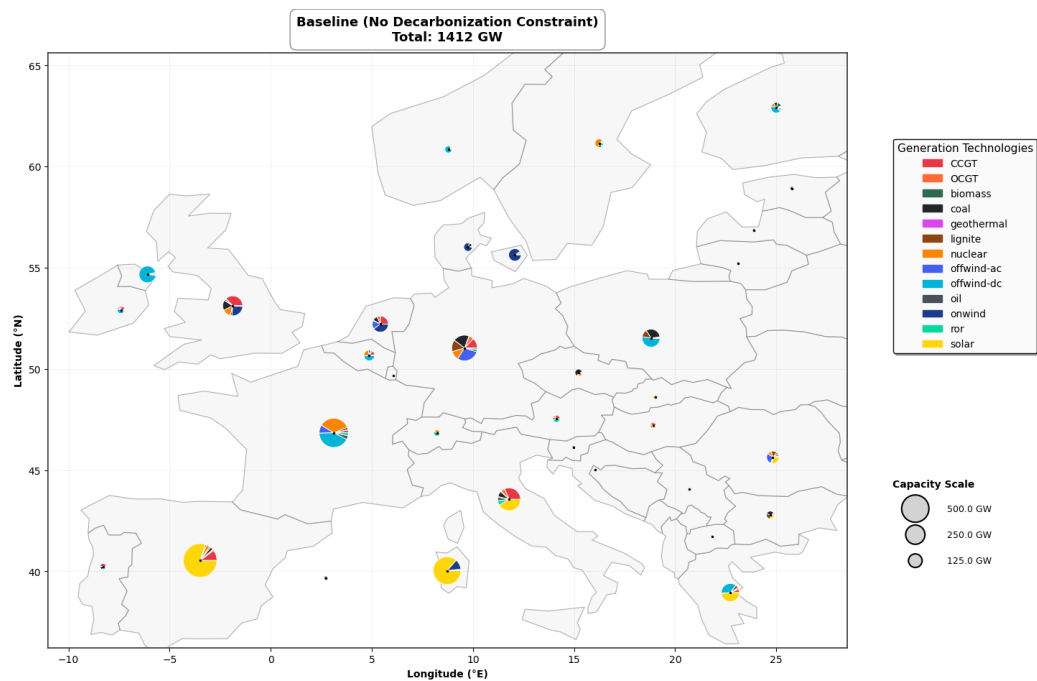


Figure 8: Figure text

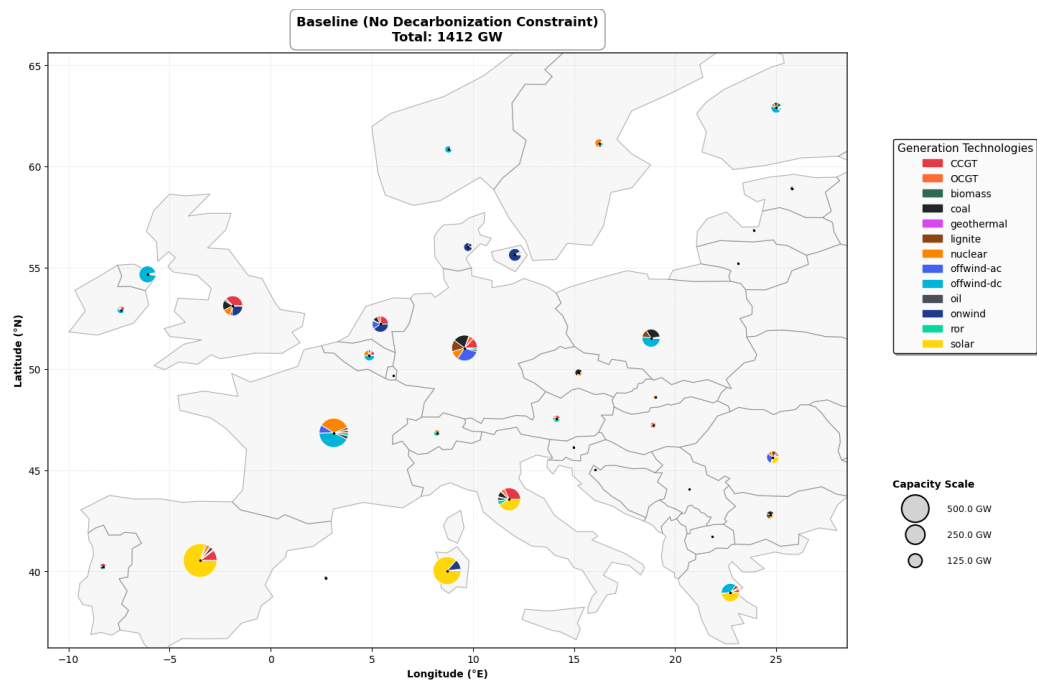


Figure 9: Figure text

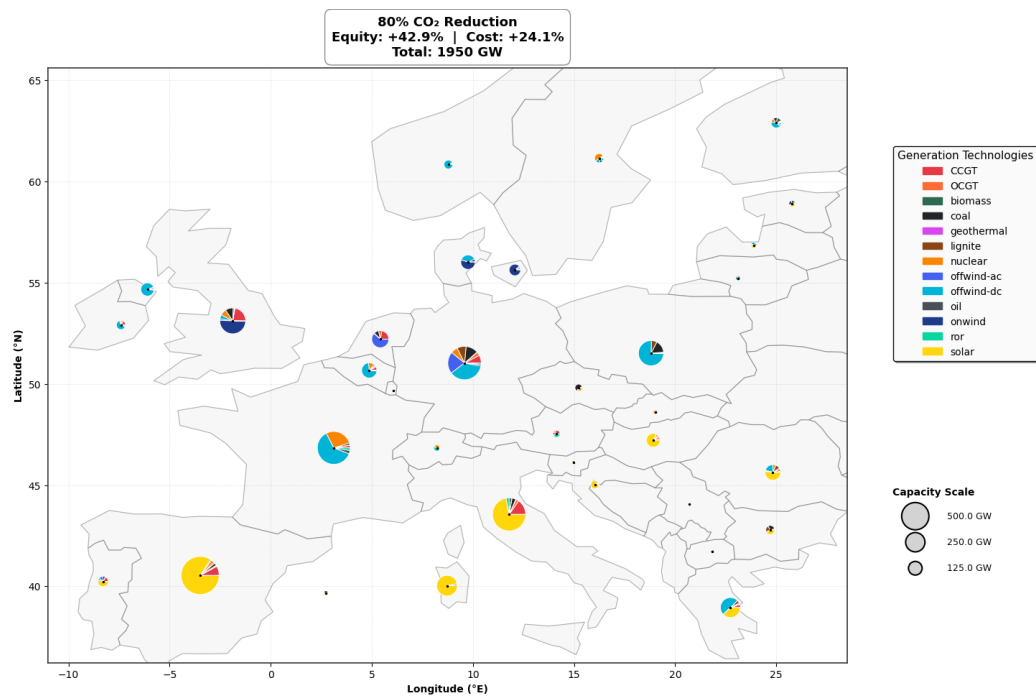


Figure 10: Figure text

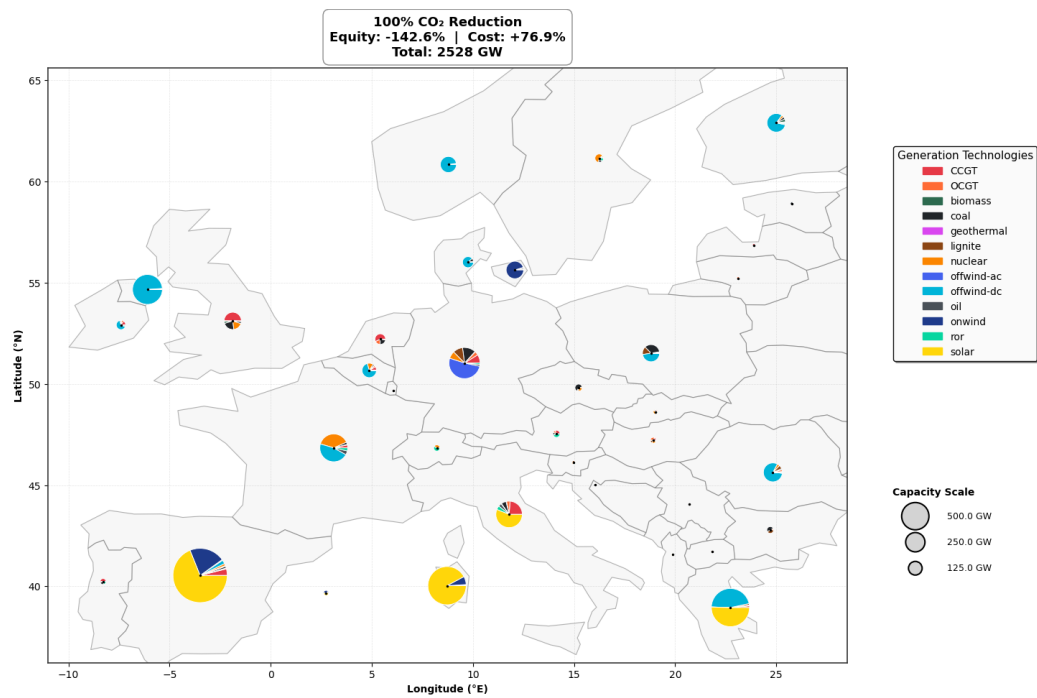


Figure 11: Figure text

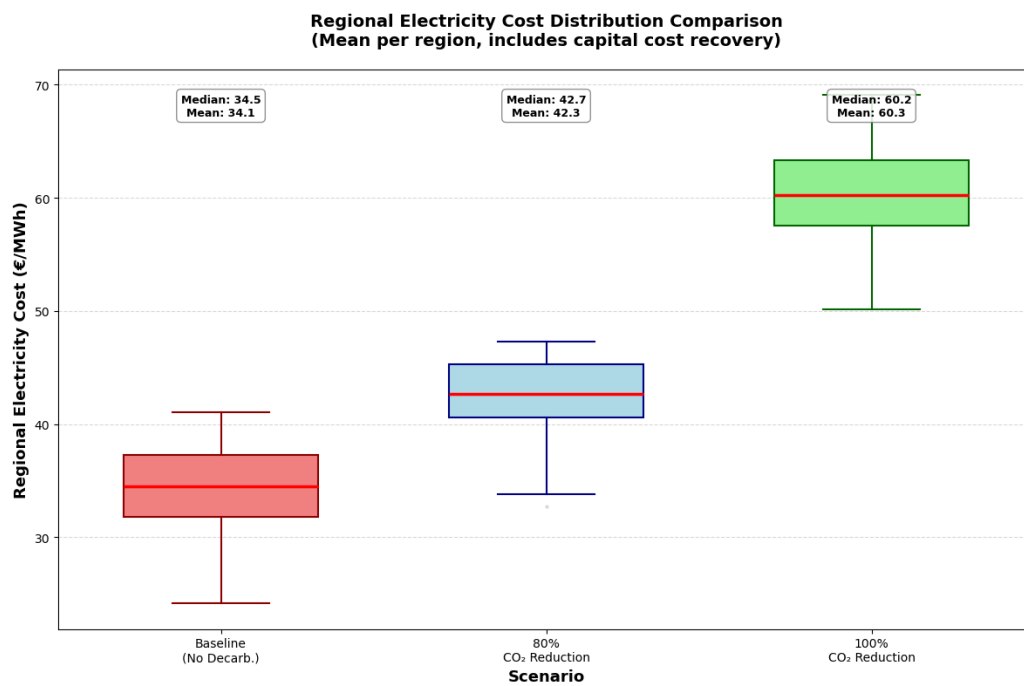


Figure 12: Figure text

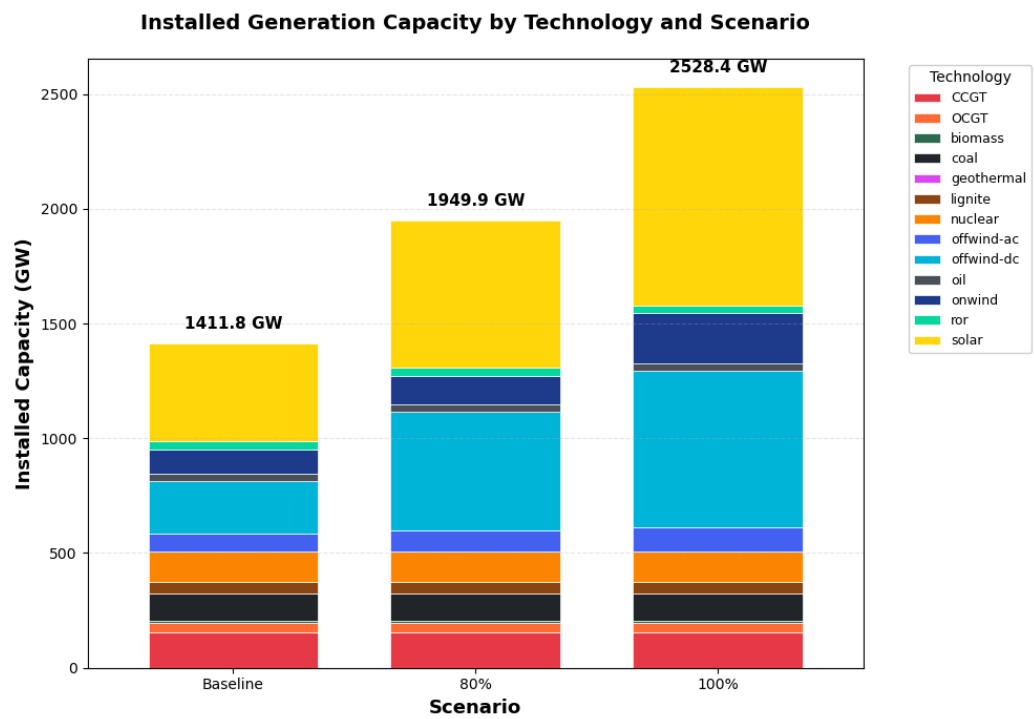


Figure 13: Figure text

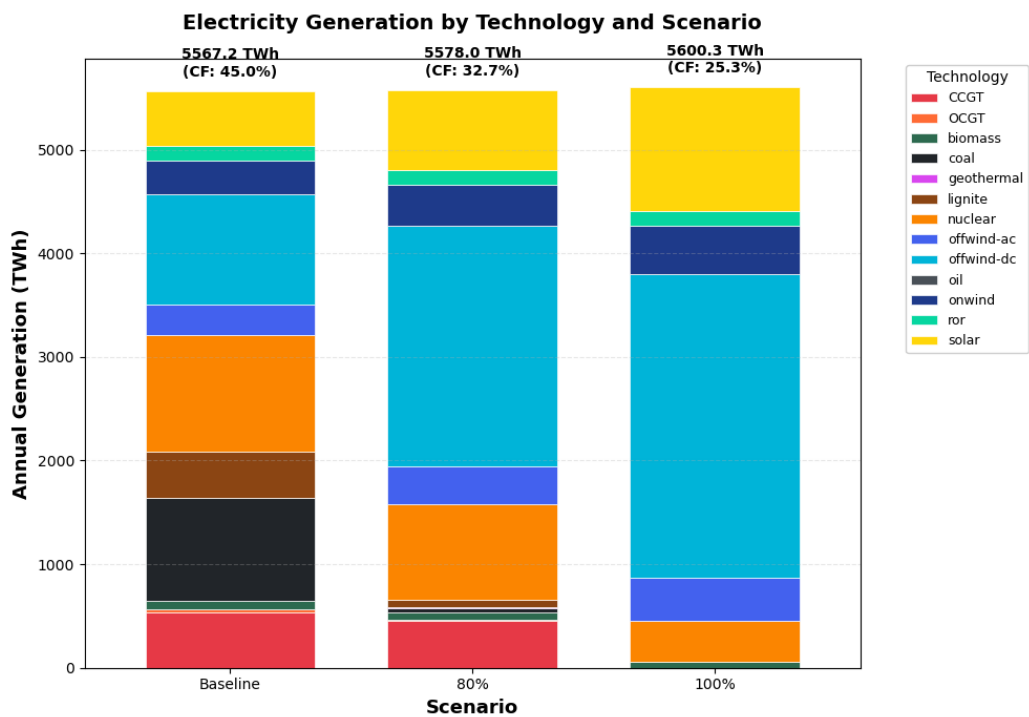


Figure 14: Figure text

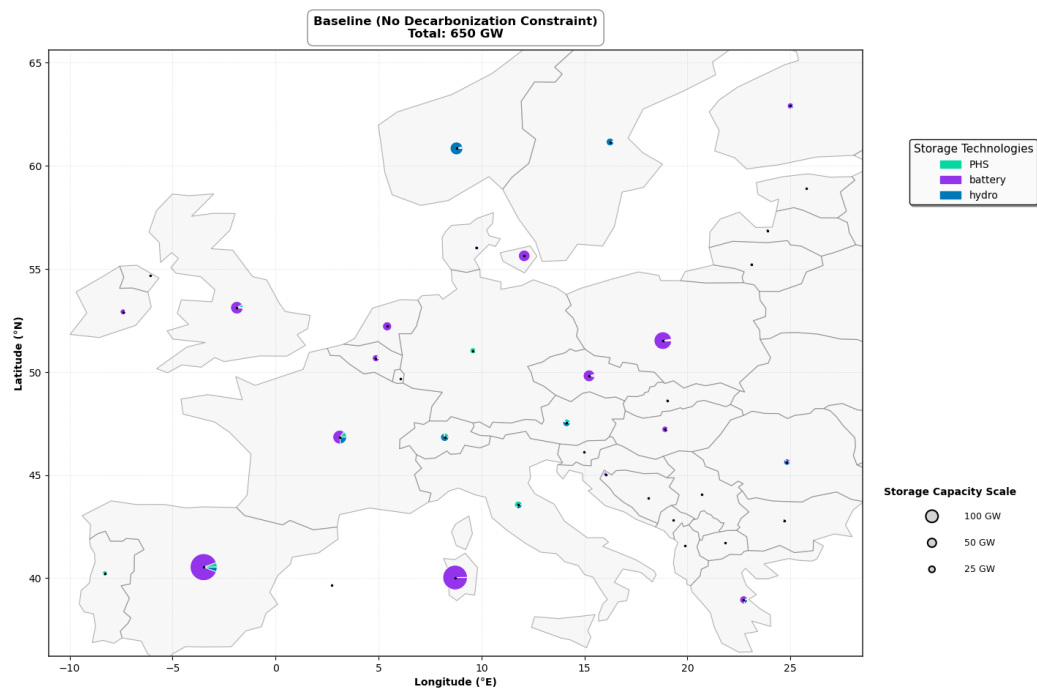


Figure 15: Figure text

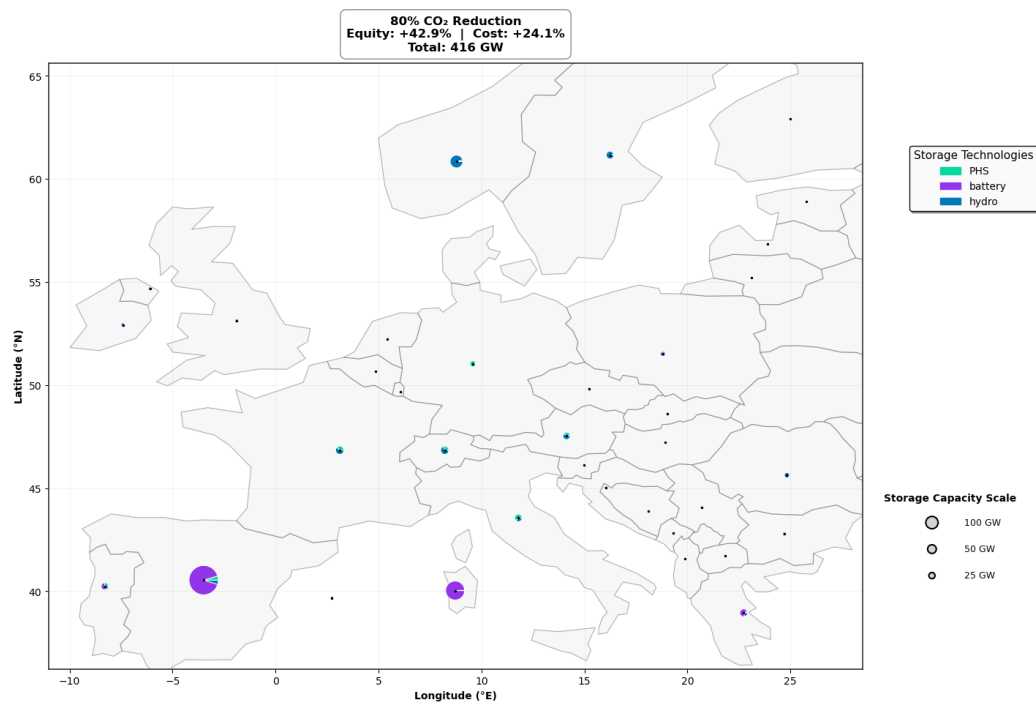


Figure 16: Figure text

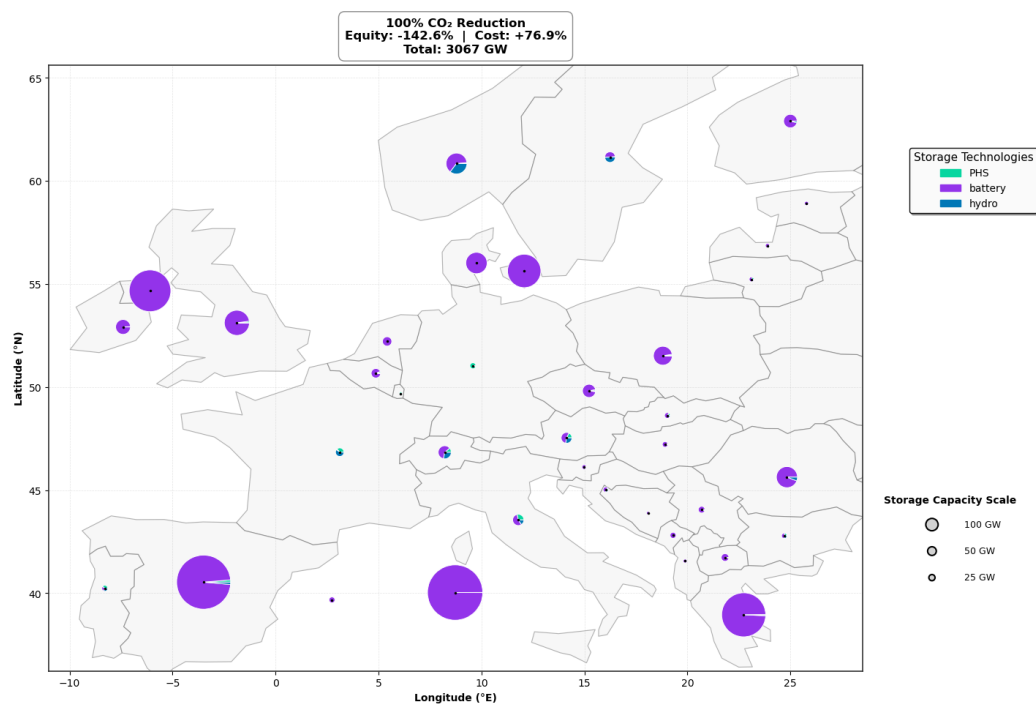


Figure 17: Figure text

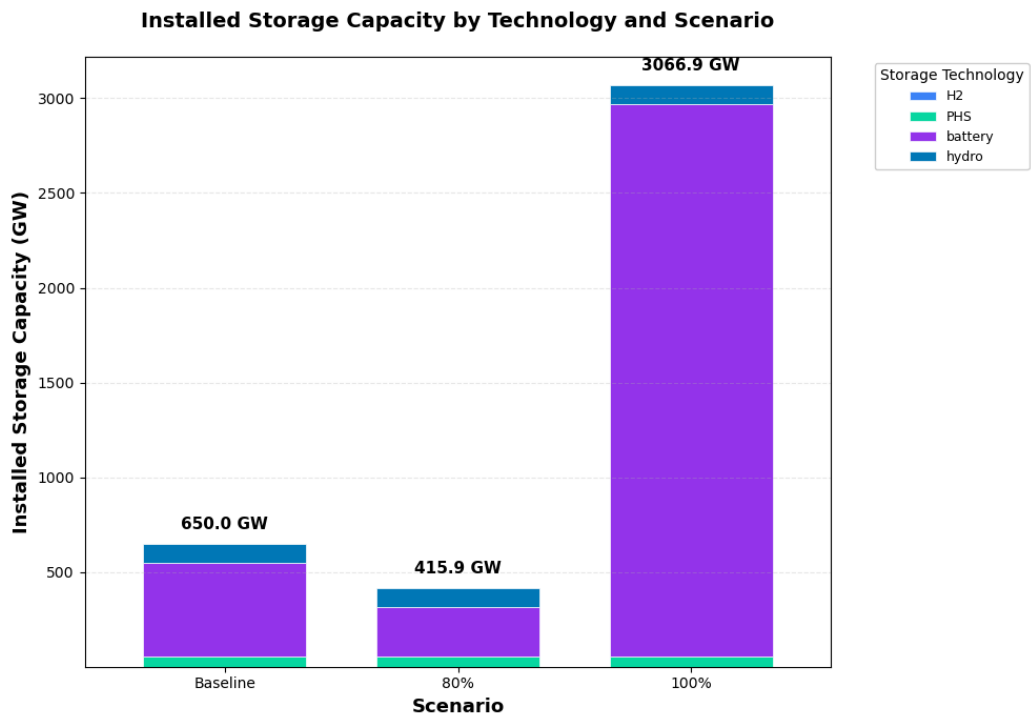


Figure 18: Figure text

4.2.1. Sanity checks and validations

Comparing systems to real systems

4.2.2. Decarbonization trends

Analyzing trends in key figures and plots for the decarbonizing systems

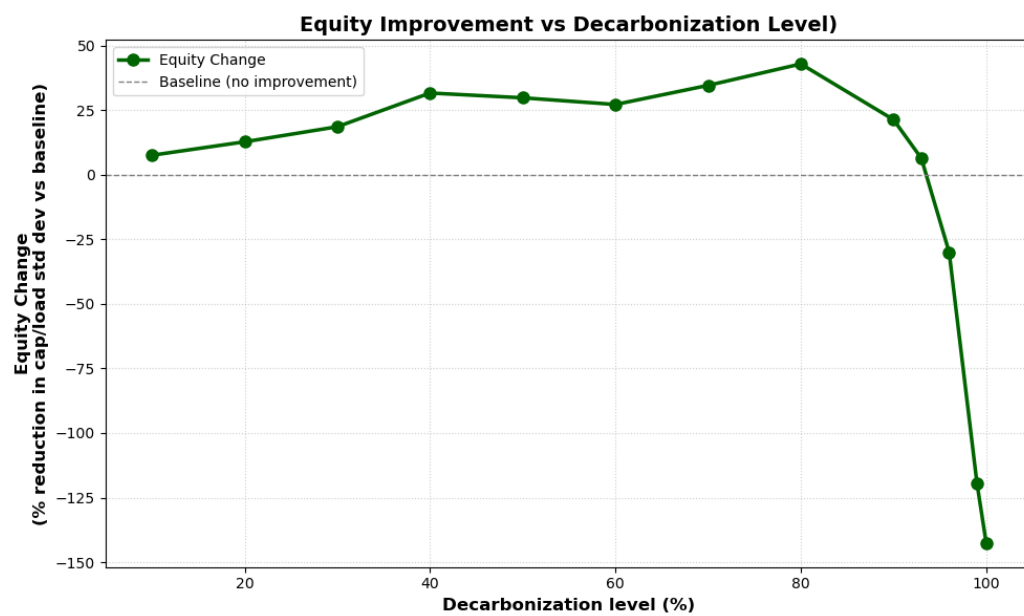


Figure 19: Figure text

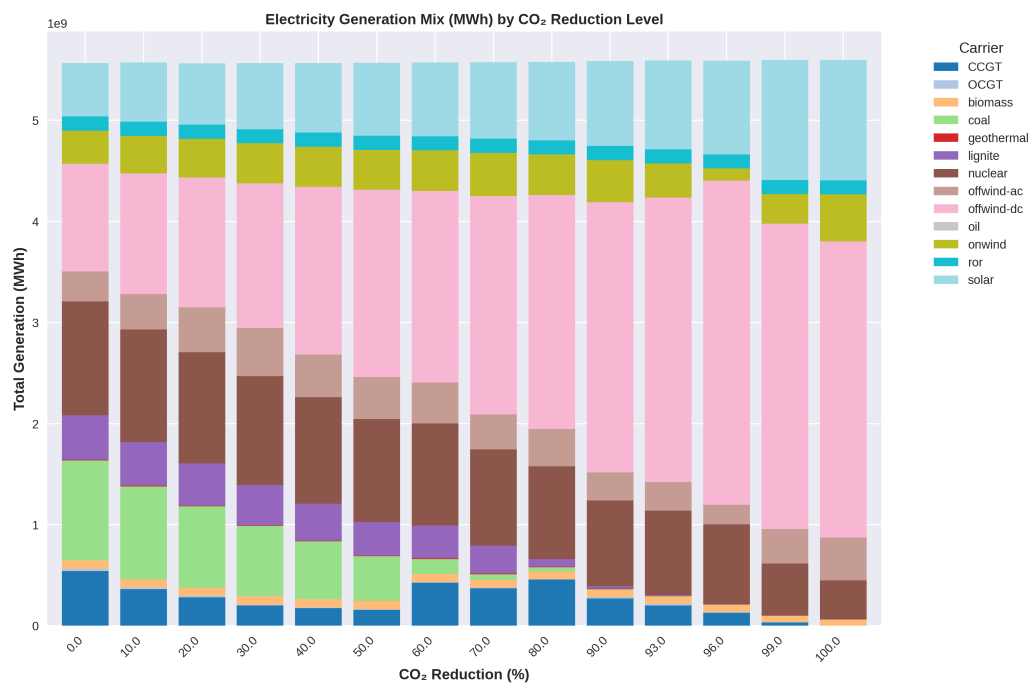


Figure 20: Figure text

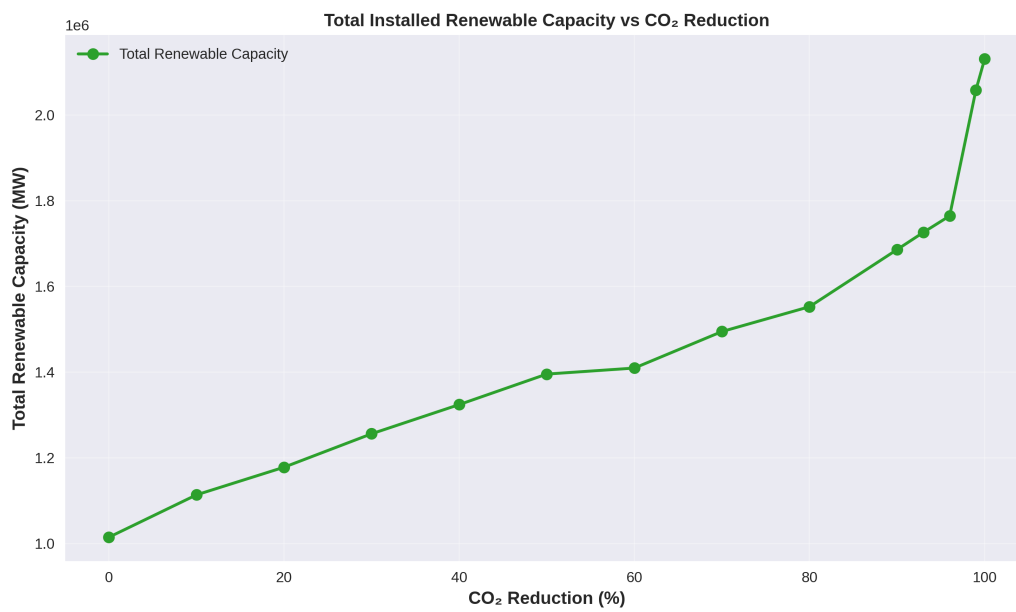


Figure 21: Figure text

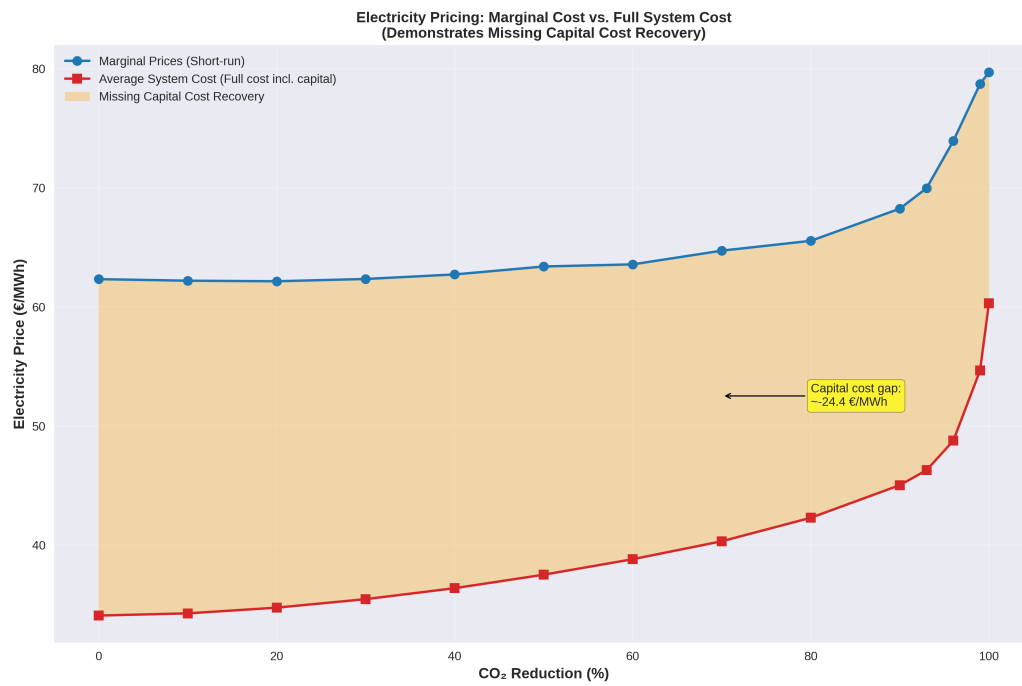


Figure 22: Figure text

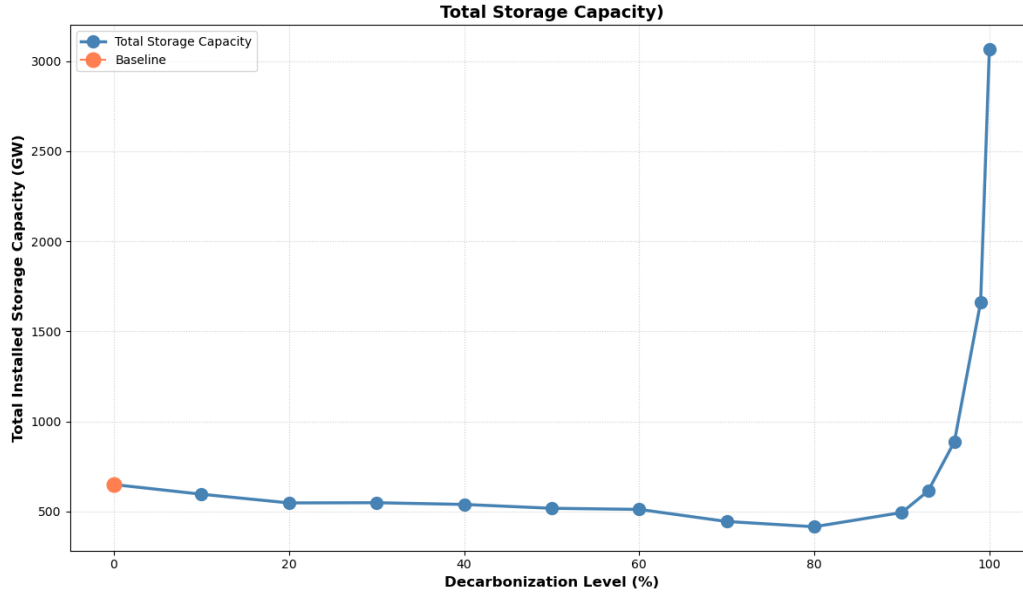


Figure 23: Figure text

4.2.3. *Driving Mechanisms in decarbonizing systems*

Discussing driving mechanisms behind trends like load-following and resource optimization

4.3. *Decentralizing systems*

Introduction to the decentralizing systems showing a few networks and comparing their topology in contrast to their baselines.

4.3.1. *Sanity checks and validations*

Comparing the key figures like marginal prices to those from real networks

4.3.2. *Decentralization trends*

Analyzing how decentralization, de-carbonization and key figures interact. Plotting 3-d response surfaces and discussing shape of solution spaces.

4.3.3. *Driving Mechanisms in decentralizing systems*

Talking about mechanisms that produce the interactions seen above

5. Discussion

Introduction to discussion

5.1. *Economic trends and societal paradigms*

What can interactions can be taken from my results and how does this align with expectations and real-world trends

5.2. *Future policy implications*

Discussing how this workflow can be used to analyze and incorporate paradigms to review implications of energy policy.

5.3. *Uses for methodology*

How else could this approach be used and how would this expand the workflows use-case.

6. Conclusion

7. Assumptions

8. Acknowledgments

The auhor would like to thank Sommer Marie Aden-Dam for her moral support and help in keeping my breaks short.

9. References

References

- [1] The Paris Agreement | UNFCCC.
URL <https://unfccc.int/process-and-meetings/the-paris-agreement>
- [2] R. A. Rodriguez, S. Becker, M. Greiner, Cost-optimal design of a simplified, highly renewable pan-European electricity system 83 658–668.
doi:10.1016/j.energy.2015.02.066.
URL <https://www.sciencedirect.com/science/article/pii/S0360544215002212>

- [3] S. Pfenninger, A. Hawkes, J. Keirstead, Energy systems modeling for twenty-first century energy challenges 33 74–86. doi:10.1016/j.rser.2014.02.003.
URL <https://www.sciencedirect.com/science/article/pii/S1364032114000872>
- [4] H. Henke, M. Dekker, F. Lombardi, R. Pietzcker, P. Fragkos, B. Zakari, R. Rodrigues, J. Sitarz, J. Emmerling, A. Fattahi, F. Dalla Longa, I. Tatarewicz, T. Fotiou, M. Lewarski, D. Huppmann, K. Kavvadias, B. Zwaan, W. Usher, Comparing energy system optimization models and integrated assessment models: Relevance for energy policy advice 3 69. doi:10.12688/openreseurope.15590.2.
- [5] E. H. Eriksen, L. J. Schwenk-Nebbe, B. Tranberg, T. Brown, M. Greiner, Optimal heterogeneity in a simplified highly renewable European electricity system 133 913–928. doi:10.1016/j.energy.2017.05.170.
URL <https://www.sciencedirect.com/science/article/pii/S0360544217309593>
- [6] The Scale of Russian Sabotage Operations Against Europe’s Critical Infrastructure.
URL <https://www.iiss.org/research-paper/2025/08/the-scale-of-russian--sabotage-operations--against-europes-critical--infrastructure/>
- [7] GNU General Public License version 2.
URL <https://opensource.org/license/gpl-2-0/>
- [8] Snakemake | Snakemake 9.14.0 documentation.
URL <https://snakemake.readthedocs.io/en/stable/>
- [9] K. Hedegaard Aden, Thesis-workflow for 'Fairness in a decarbonizing Europe'.
URL <https://github.com/KrisAden/Thesis-workflow>
- [10] GNU General Public License v2.0 - GNU Project - Free Software Foundation.
URL <https://www.gnu.org/licenses/old-licenses/gpl-2.0.en.html>

- [11] T. Brown, J. Hörsch, D. Schlachtberger, PyPSA: Python for Power System Analysis, *Journal of Open Research Software* 6 (4) (2018). arXiv:1707.09913, doi:10.5334/jors.188.
URL <https://doi.org/10.5334/jors.188>
- [12] J. Hoersch, F. Hofmann, D. Schlachtberger, T. Brown, Pypsa-eur: An open optimisation model of the european transmission system, *Energy Strategy Reviews* 22 (2018) 207–215. arXiv:1806.01613, doi:10.1016/j.esr.2018.08.012.
- [13] J. Hörsch, F. Neumann, F. Hofmann, D. Schlachtberger, M. Frysztacki, J. Hampp, P. Glaum, T. Brown, PyPSA-Eur: An Open Optimisation Model of the European Transmission System (Dataset). doi:10.5281/zenodo.7646728.
URL <https://zenodo.org/records/7646728>
- [14] Transparency Platform.
URL <https://transparency.entsoe.eu/load/total/dayAhead?appState=%7B%22sa%22%3A%5B%22CTY%7C10YBE-----2%22%5D%2C%22st%22%3A%22CTY%22%2C%22mm%22%3Atrue%2C%22ma%22%3Afalse%2C%22sp%22%3A%22HALF%22%2C%22dt%22%3A%22TABLE%22%2C%22df%22%3A%5B%222025-12-01%22%2C%222025-12-01%22%5D%2C%22tz%22%3A%22CET%22%7D>
- [15] Carbon Free Europe.
URL <https://www.carbonfreeeurope.org>
- [16] Technology Data for Generation of Electricity and District Heating | Energistyrelsen.
URL <https://ens.dk/en/analyses-and-statistics/technology-data-generation-electricity-and-district-heating>
- [17] The Future of Geothermal Energy – Analysis.
URL <https://www.iea.org/reports/the-future-of-geothermal-energy>
- [18] Projected Costs of Generating Electricity.
URL https://www.oecd-nea.org/jcms/pl_28612/projected-costs-of-generating-electricity

- [19] Technology Data for Energy Storage | Energistyrelsen.
URL <https://ens.dk/en/analyses-and-statistics/technology-data-energy-storage>
- [20] UCloud User Guide — UCloud.
URL <https://docs.cloud.sdu.dk/index.html>
- [21] Barrier Logging - Gurobi Optimizer Reference Manual.
URL <https://docs.gurobi.com/projects/optimizer/en/current/concepts/logging/barrier.html>
- [22] Energy Balance - Documentation.
URL <https://docs.pypsa.org/latest/user-guide/optimization/energy-balance/>
- [23] T. T. Pedersen, M. Victoria, M. G. Rasmussen, G. B. Andresen, Modeling all alternative solutions for highly renewable energy systems 234 121294. doi:10.1016/j.energy.2021.121294.
URL <https://www.sciencedirect.com/science/article/pii/S0360544221015425>
- [24] A. Kies, Deriving the zero-profit condition for pypsa, unpublished internal document.

10. Appendices

10.1. Appendix A: Generator cost assumptions

10.2. Appendix B: Storage cost assumptions

This complicates scarcity rents as hydro does not have a capital cost defined

Table 3: Generator technology cost assumptions used in the optimization model.

Technology	Capital Cost (EUR/MW/a)	Marginal Cost (EUR/MWh)
CCGT	99.0k	47.20
OCGT	45.1k	58.38
biomass	379.2k	14.96
coal	180.2k	24.10
geothermal	490.7k	0.00
lignite	180.2k	13.49
nuclear	535.5k	16.90
offwind-ac	196.7k	0.01
offwind-dc	219.4k	0.01
oil	37.9k	130.23
onwind	133.8k	0.01
ror	270.9k	0.00
solar	55.1k	0.01

Table 4: Storage technology cost assumptions extracted from the optimized network.

Technology	Capital Cost (EUR/MW/a)	Marginal Cost (EUR/MWh)
Battery	13.7k	0.00
Hydrogen	205.0k	0.00
PHS	160.6k	0.00
Hydro	0.0k	0.00